

ECE 468: Lab 3 Report

UNIVERSITY OF
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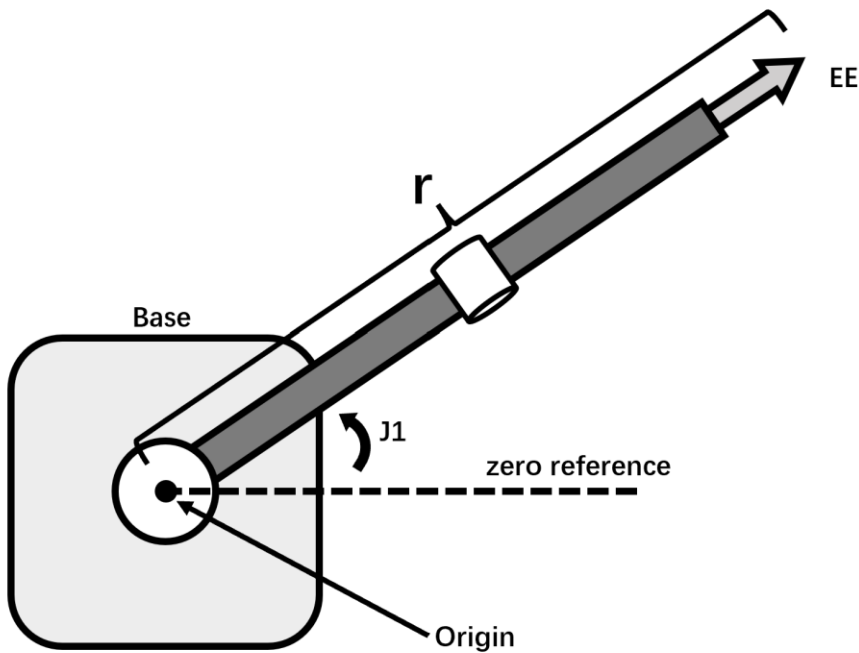
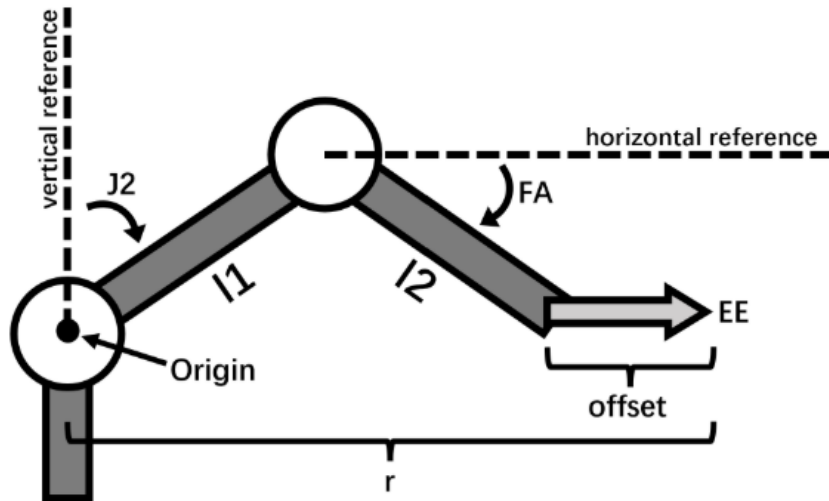
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The Derivation (20 marks)

IK Diagram (5 marks)



Inverse Kinematics Derivation (10 marks)

We derivation our inverse kinematic (IK) based on forward kinematic (FK) formula. Given (J_1, J_2, FA) , we can use FK to calculate (x, y, z) by

$$\begin{aligned}z &= l_1 \cos(J_2) - l_2 \sin(FA) \\r &= l_1 \sin(J_2) + l_2 \cos(FA) + EE_o \\x &= r \cos(J_1) \\y &= r \sin(J_1)\end{aligned}$$

where l_1, l_2, EE_o are some known parameters of the manipulator.

Therefore, we have

$$\frac{y}{x} = \frac{r \sin(J_1)}{r \cos(J_1)} = \tan(J_1) \Rightarrow J_1 = \arctan 2(y, x)$$

Note $\arctan(\cdot)$ will have range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, and we have $J_1 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ based on the manipulator design, so the formula is valid. In our code, we use $\arctan 2(\cdot, \cdot)$ function, which is also identify the quadrant of the angle.

We also have

$$\begin{aligned}x^2 + y^2 &= r^2 \cos^2(J_1) + r^2 \sin^2(J_1) = r^2(\cos^2(J_1) + \sin^2(J_1)) = r^2 \\ \Rightarrow r &= \sqrt{x^2 + y^2} = l_1 \sin(J_2) + l_2 \cos(FA) + EE_o\end{aligned}$$

Let $r_0 = r - EE_o = l_1 \sin(J_2) + l_2 \cos(FA)$, then we will have following system of equations of J_2 and FA

$$\begin{aligned}l_1 \sin(J_2) + l_2 \cos(FA) &= r_0 \\ l_1 \cos(J_2) - l_2 \sin(FA) &= z\end{aligned}$$

where l_1, l_2, r_0, z are all known constants, and we also know $J_2, FA \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ based on the designing of manipulator.

According to the query, system of equations with such form

$$\begin{aligned}C_1 \sin(\theta_1) + C_2 \cos(\theta_2) &= B_1 \\ C_1 \cos(\theta_1) - C_2 \sin(\theta_2) &= B_2\end{aligned}$$

and with constraints $\theta_1, \theta_2 \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, has the analytical solution formula

$$\begin{aligned}\theta_1 &= \arctan 2(B_1, B_2) - \arctan 2(Y, X) \\ \theta_2 &= \theta_1 - \gamma\end{aligned}$$

where

$$\gamma = \arcsin\left(\frac{B_1^2 + B_2^2 - C_1^2 - C_2^2}{2C_1C_2}\right)$$

$$X = C_1 + C_2 \sin \gamma$$

$$Y = C_2 \cos \gamma$$

Therefore, we only need to apply the formula by use $C_1 = l_1$, $C_2 = l_2$, $\theta_1 = J_2$, $\theta_2 = FA$, $B_1 = r_0$, $B_2 = z$, then we will get

$$J_2 = \arctan 2(r_0, z) - \arctan 2(Y, X)$$

$$FA = J_2 - \gamma$$

where

$$\gamma = \arcsin\left(\frac{r_0^2 + z^2 - l_1^2 - l_2^2}{2l_1l_2}\right)$$

$$X = l_1 + l_2 \sin \gamma$$

$$Y = l_2 \cos \gamma$$

In addition, we search and write the approximated proof of the analytical solution formula

$$C_1 \sin(\theta_1) + C_2 \cos(\theta_2) = B_1$$

$$C_1 \cos(\theta_1) - C_2 \sin(\theta_2) = B_2$$

Square two equations

$$B_1^2 = C_1^2 \sin^2(\theta_1) + 2C_1C_2 \sin(\theta_1) \cos(\theta_2) + C_2^2 \cos^2(\theta_2)$$

$$B_2^2 = C_1^2 \cos^2(\theta_1) - 2C_1C_2 \cos(\theta_1) \sin(\theta_2) + C_2^2 \sin^2(\theta_2)$$

Sum two squared equations

$$B_1^2 + B_2^2$$

$$= C_1^2 \sin^2(\theta_1) + 2C_1C_2 \sin(\theta_1) \cos(\theta_2) + C_2^2 \cos^2(\theta_2) + C_1^2 \cos^2(\theta_1) - 2C_1C_2 \cos(\theta_1) \sin(\theta_2) + C_2^2 \sin^2(\theta_2)$$

$$= C_1^2 (\sin^2(\theta_1) + \cos^2(\theta_1)) + C_2^2 (\sin^2(\theta_2) + \cos^2(\theta_2)) + 2C_1C_2 (\sin(\theta_1) \cos(\theta_2) - \cos(\theta_1) \sin(\theta_2))$$

$$= C_1^2 + C_2^2 + 2C_1C_2 \sin(\theta_1 - \theta_2)$$

$$\Rightarrow \sin(\theta_1 - \theta_2) = \frac{B_1^2 + B_2^2 - C_1^2 - C_2^2}{2C_1C_2}$$

$$\text{Let } \gamma = \theta_1 - \theta_2 \Rightarrow \theta_2 = \theta_1 - \gamma, \text{ and } \gamma = \arcsin\left(\frac{B_1^2 + B_2^2 - C_1^2 - C_2^2}{2C_1C_2}\right)$$

$$C_1 \sin(\theta_1) + C_2 \cos(\theta_1 - \gamma) = B_1 \quad (1)$$

$$C_1 \cos(\theta_1) - C_2 \sin(\theta_1 - \gamma) = B_2 \quad (2)$$

$$\textcircled{1} \Rightarrow C_1 \sin(\theta_1) + C_2(\cos(\theta_1) \cos \gamma + \sin(\theta_1) \sin \gamma) = B_1$$

$$\Rightarrow (C_1 + C_2 \sin \gamma) \sin(\theta_1) + (C_2 \cos \gamma) \cos(\theta_1) = B_1$$

$$\textcircled{2} \Rightarrow C_1 \cos(\theta_1) - C_2(\sin(\theta_1) \cos \gamma - \cos(\theta_1) \sin \gamma) = B_2$$

$$\Rightarrow (C_1 + C_2 \sin \gamma) \cos(\theta_1) - (C_2 \cos \gamma) \sin(\theta_1) = B_2$$

Let $X = C_1 + C_2 \sin \gamma$ and $Y = C_2 \cos \gamma$, we can simplify to

$$X \sin(\theta_1) + Y \cos(\theta_1) = B_1$$

$$X \cos(\theta_1) - Y \sin(\theta_1) = B_2$$

which is a linear equation to $\sin(\theta_1)$ and $\cos(\theta_1)$, solve to get

$$\sin(\theta_1) = \frac{XB_1 - YB_2}{X^2 + Y^2}$$

$$\cos(\theta_1) = \frac{YB_1 + XB_2}{X^2 + Y^2}$$

$$\Rightarrow \tan(\theta_1) = \frac{\sin(\theta_1)}{\cos(\theta_1)} = \frac{\left(\frac{XB_1 - YB_2}{X^2 + Y^2}\right)}{\left(\frac{YB_1 + XB_2}{X^2 + Y^2}\right)} = \frac{XB_1 - YB_2}{YB_1 + XB_2}$$

Divide both up and down by XB_2 , we will have

$$\tan(\theta_1) = \frac{\frac{XB_1}{XB_2} - \frac{YB_2}{XB_2}}{\frac{YB_1}{XB_2} + \frac{XB_2}{XB_2}} = \frac{\frac{B_1}{B_2} - \frac{Y}{X}}{\frac{Y}{X} \frac{B_1}{B_2} + 1}$$

Note $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{\tan \alpha \tan \beta + 1}$, then if let $\tan \alpha = \frac{B_1}{B_2}$ and $\tan \beta = \frac{Y}{X}$, we can convert our formula to

$$\tan(\theta_1) = \frac{\tan \alpha - \tan \beta}{\tan \alpha \tan \beta + 1} \text{ where } \theta_1 = \alpha - \beta$$

Therefore, we have

$$\theta_1 = \arctan 2(B_1, B_2) - \arctan 2(Y, X)$$

$$\theta_2 = \theta_1 - \gamma$$

Offline Validation (5 marks)

We are given the txt file with several rows, and each row represent a point, and 6 columns, which each column represent x, y, z, J_1, J_2, FA for a point respectively. We write the code to read the text file, and for each point, we use IK function we write to calculate the joint angles, and compare it with the real joint angles and calculate the error by Euclid distance.

$$q_{IK} = (J_{1IK}, J_{2IK}, FA_{IK}) = IK(x, y, z)$$

$$q = (J_1, J_2, FA)$$

$$e = ||q_{IK} - q|| = \sqrt{(J_{1IK} - J_1)^2 + (J_{2IK} - J_2)^2 + (FA_{IK} - FA)^2}$$

After validation, all the errors are within 10^{-5} degree level, which proves the IK function gives desired results.

Mean Error	6.137×10^{-6}
Max Error	2.519×10^{-5}
Min Error	2.313×10^{-7}

Testing (10 marks)

Testing Procedure and Results (10 marks)

We selected several representative points distributed in the manipulator's workspace in joint angle coordinate $q_{test} = (J_1, J_2, FA)$.

After controlling the manipulator to move to each point, we output the Cartesian coordinates measured by the manipulator $p_{meas} = (x_{meas}, y_{meas}, z_{meas})$.

Then, using Cartesian coordinates as input, put them into our IK function written before, and calculate the IK joint angles

$$q_{IK} = IK(x_{meas}, y_{meas}, z_{meas}) = (J_{1IK}, J_{2IK}, FA_{IK}).$$

Finally, calculate the error using Euclid distance as

$$e = ||q_{IK} - q_{test}|| = \sqrt{(J_{1IK} - J_1)^2 + (J_{2IK} - J_2)^2 + (FA_{IK} - FA)^2}$$

After testing, the following results were obtained

Test Joint (deg)	IK Joint (deg)	Error (deg)
[0, 20, 70]	[0.00000000, 20.00000461, 70.00000712]	0.00000848
[45, 45, 45]	[45.00000239, 44.99999258, 45.00000765]	0.00001092
[-45, 30, 70]	[-45.00000174, 30.00000103, 70.00001189]	0.00001207
[0, 30, 60]	[0.00000000, 29.99999493, 60.00000358]	0.00000621
[20, 30, 60]	[20.00000038, 29.99999672, 60.00000193]	0.00000382
[-20, 30, 60]	[-20.00000038, 29.99999672, 60.00000193]	0.00000382

After calculation, the mean error is 0.00000755, with max error up to 0.00001207, and min error down to 0.00000382. The experimental results indicate that the inverse kinematics algorithm remains highly accurate during practical testing.

Camera Calibration (15 marks)

First, we calibrated the camera by taking multiple photos for graph paper, and generating a camera matrix and dist coefficients.

$$C = \begin{pmatrix} 1503.80025 & 0 & 313.485612 \\ 0 & 33158.98471 & 240.184062 \\ 0 & 0 & 1 \end{pmatrix}$$

$$D = \begin{pmatrix} 0.204070743 \\ -4.03613477 \\ -0.0108718487 \\ -0.0010776246 \\ -19.3417151 \end{pmatrix}$$

After calibrated, during the whole process, the camera and manipulator never been touched again, in order to prevent any error caused by shifting the base.

We input those parameters to our code template, put the ArUco sticker on the end effect of the manipulator, and then use 10 points to calculate the linear transformation matrix R and translation vector T .

$$R = \begin{pmatrix} -0.60740703 & -0.55410305 & 0.56923326 \\ -0.72715913 & 0.0993171 & -0.67924643 \\ 0.31983792 & -0.82650222 & -0.463247 \end{pmatrix}$$

$$T = \begin{pmatrix} -0.23747033 \\ 0.62191486 \\ 0.34901836 \end{pmatrix}$$

After that, we use 20 new points for validation. After controlling manipulator to a certain cartesian coordinate $p = (x, y, z)^T$, the camera will capture a coordinate $p_{cm} = (x_{cm}, y_{cm}, z_{cm})^T$. We will use the linear transformation to map the camera coordinate to real coordinate by

$$p_{tran} = Rp_{cm} + T = (x_{tran}, y_{tran}, z_{tran})^T$$

Then, we will calculate the error by Euclid distance

$$e = ||p_{tran} - p|| = \sqrt{(x_{tran} - x)^2 + (y_{tran} - y)^2 + (z_{tran} - z)^2}$$

After validation, following results were obtained

Real Coordinate (mm)	Predicted Coordinate (mm)	Error (mm)
[160, -30, -10]	[139.92, -2.90, 14.06]	41.43
[180, -30, -10]	[162.57, -11.27, 7.69]	31.11
[210, -30, -10]	[188.55, -12.73, 5.00]	31.36
[240, -30, -10]	[216.34, -16.21, 1.06]	29.53

[160, 20, -10]	[150.25, 50.64, -0.66]	33.49
[180, 20, -10]	[169.01, 47.36, -3.57]	30.18
[210, 20, -10]	[195.66, 43.79, -6.75]	27.96
[240, 20, -10]	[222.57, 40.17, -10.18]	26.66
[160, 60, -10]	[155.08, 97.21, -9.33]	37.54
[180, 60, -10]	[169.96, 97.93, -9.02]	39.24
[210, 60, -10]	[198.46, 91.73, -13.95]	34.00
[240, 60, -10]	[226.72, 85.97, -18.54]	30.39
[250, 20, -20]	[245.34, 21.63, -24.66]	6.78
[200, -20, -20]	[200.18, -25.34, -13.29]	8.57
[230, -20, -20]	[222.79, -22.67, -13.72]	9.93
[170, 30, -20]	[172.80, 45.10, -16.18]	15.83
[200, 30, -20]	[196.49, 45.97, -16.76]	16.67
[230, 30, -20]	[223.88, 41.15, -20.80]	12.74
[180, 50, -30]	[193.47, 55.40, -30.94]	14.55
[220, 50, -30]	[227.53, 52.16, -34.36]	8.97

According to the calculation, the mean error is 24.35 mm, with max error up to 41.43 mm, and min error down to 6.78 mm. Dr. Cooper indicate that this error is within the normal range. The remaining error mainly originates from camera calibration inaccuracies, the mounting offset between the ArUco marker and the robot end-effector, and the finite positioning accuracy of the Dobot manipulator, all of which introduce systematic and measurement errors into the estimated robot coordinates.

For the error correction, we can first calculate the mean error in vector, which is

$$b = \frac{1}{N} \sum_{i=1}^N (p_i - p_{tran_i}) \approx (8.62, -17.75, -4.26)^T$$

We can see $b \neq 0$, which means the prediction error contains a systematic bias, and we can optimize the error by shifting all the data use $p_{corr} = p_{tran} + b$.

Real Coord (mm)	Predicted Coord (mm)	Corrected Coord (mm)	Corrected Error (mm)
[160, -30, -10]	[139.92, -2.90, 14.06]	[148.54, -20.65, 9.8]	24.72
[180, -30, -10]	[162.57, -11.27, 7.69]	[171.19, -29.02, 3.43]	16.09
[210, -30, -10]	[188.55, -12.73, 5.00]	[197.17, -30.48, 0.74]	16.74
[240, -30, -10]	[216.34, -16.21, 1.06]	[224.96, -33.96, -3.2]	16.98
[160, 20, -10]	[150.25, 50.64, -0.66]	[158.87, 32.89, -4.92]	13.90
[180, 20, -10]	[169.01, 47.36, -3.57]	[177.63, 29.61, -7.83]	10.13
[210, 20, -10]	[195.66, 43.79, -6.75]	[204.28, 26.04, -11.01]	8.37
[240, 20, -10]	[222.57, 40.17, -10.18]	[231.19, 22.42, -14.44]	10.15
[160, 60, -10]	[155.08, 97.21, -9.33]	[163.7, 79.46, -13.59]	20.13
[180, 60, -10]	[169.96, 97.93, -9.02]	[178.58, 80.18, -13.28]	20.49
[210, 60, -10]	[198.46, 91.73, -13.95]	[207.08, 73.98, -18.2]	16.47
[240, 60, -10]	[226.72, 85.97, -18.54]	[235.34, 68.22, -22.8]	15.90

[250, 20, -20]	[245.34, 21.63, -24.66]	[253.96, 3.88, -28.91]	18.85
[200, -20, -20]	[200.18, -25.34, -13.29]	[208.8, -43.09, -17.55]	24.84
[230, -20, -20]	[222.79, -22.67, -13.72]	[231.41, -40.42, -17.98]	20.57
[170, 30, -20]	[172.80, 45.10, -16.18]	[181.42, 27.35, -20.44]	11.73
[200, 30, -20]	[196.49, 45.97, -16.76]	[205.11, 28.22, -21.02]	5.51
[230, 30, -20]	[223.88, 41.15, -20.80]	[232.5, 23.4, -25.05]	8.69
[180, 50, -30]	[193.47, 55.40, -30.94]	[202.09, 37.65, -35.2]	25.84
[220, 50, -30]	[227.53, 52.16, -34.36]	[236.15, 34.41, -38.62]	24.05

After correction, the mean error drop to 16.51 mm, with max error 25.84 mm, and min error 5.51 mm. Compared with the raw error, the corrected error is significantly reduced, demonstrating that a large portion of the prediction error originates from a systematic bias rather than random noise. The remaining error is mainly attributed to residual camera calibration inaccuracies, ArUco pose estimation uncertainty, and the limited repeatability of the robot manipulator.